

Anil Hirpara

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Academic Details

Course	Board/University	School/College	Year of passing	Percentage/CGPA
Ph.D. (Physics)	Sardar Patel University	Department of Physics, SPU, Vallabh Vidyanagar.	October-2023	N.A.
M.Sc. Physics	Sardar Patel University	Shri Alpesh N Patel Post Graduate Institute of Science and Research, Anand.	April - 2017	8.2
B.Sc. Physics	Veer Narmad South Gujarat University	Navyug Science College, Surat.	April-2015	6.87
H.S.C	GSHSEB	K. K. Parekh & Maheta R P Vidyalay High School, Amreli.	March-2012	52 %
S.S.C	GSEB	Shri Madhyamik shala, Nariyeli-moli.	March-2010	73.54 %

Other Exams

Course	Board/University	Subject	Year of passing
Gujarat State Eligibility Test	The Maharaj Saiyajirao University of Baroda (MSU Baroda)	Physics	December - 2018
CSIR - UGC National Eligibility Test	The Council of Scientific & Industrial Research (CSIR)	Physics	June - 2018
Course on Computer Concepts (CCC)	National Institute of Electronics & Information Technology (NIELIT)	Certificate in Computer Concepts	November - 2019

Experience

- Assisted in M.Sc. Semester (1 & 2) Laboratory courses at Department of Physics, Sardar Patel University, Vallabh Vidyanagar during the year 2019-2020.
- Assistant Professor at Indian Institute of Teacher Education, Gandhinagar for academic year 2023-2024.

Awards And Achievements

- 2017: Secured **University 1st rank out 147 students in M.Sc. (Physics)** at Sardar Patel University.
- 2018: Qualified **Junior Research Fellowship (All India rank 138)** conducted by The Council of Scientific & Industrial Research (CSIR) June - **2018**.
- 2019: Selected for **INSPIRE fellowship** by Department of Science and Technology, India.
- 2021: Fellowship designation upgraded from **Junior Research Fellowship to Senior Research Fellowship**.

Research Publications

1. **A.B. Hirpara**, S.H. Chaki, A.J. Khimani, R.M. Kannaujiya, M.P. Deshpande, Thermal Investigation of Direct Vapor Transport (DVT) Grown Quaternary Cu₂ZnSnS₄ Single Crystals, *Int. J. Thermophys.* 42 (2021) 1–15. <https://doi.org/10.1007/s10765-020-02772-y>.
2. **A.B. Hirpara**, S.H. Chaki, A.J. Khimani, R.M. Kannaujiya, M.P. Deshpande, Anisotropic photoresponse investigation of chemical vapor transport (CVT) grown quaternary Cu₂ZnSnS₄ single crystals, *J. Mater. Sci. Mater. Electron.* 33 (2022) 20303–20316. <https://doi.org/10.1007/s10854-022-08846-x>.
3. **A.B. Hirpara**, S.H. Chaki, R.M. Kannaujiya, A.J. Khimani, Z.R. Parekh, Y.H. Vaidya, R.K. Giri, M.P. Deshpande, Biological investigation of sonochemically synthesized CZTS nanoparticles, *Appl. Surf. Sci. Adv.* 12 (2022) 100338. <https://doi.org/10.1016/j.apsadv.2022.100338>.
4. **A.B. Hirpara**, S.H. Chaki, R.M. Kannaujiya, M.P. Deshpande, Photoresponse application of the dip - coated - Cu₂ZnSnS₄ thin film, *Appl. Phys. A.* (2023). <https://doi.org/10.1007/s00339-023-06507-2>.
5. A.J. Khimani, S.H. Chaki, **A.B. Hirpara**, R. V. Patel, M.P. Deshpande, Enhancing photo-detection properties of Sb_{0.15}Sn_{0.85}S₂ alloy, *Mater. Res. Express.* 6 (2019) 96303. <https://doi.org/10.1088/2053-1591/ab2cea>.
6. R.M. Kannaujiya, A.J. Khimani, S.H. Chaki, S.M. Chauhan, **A.B. Hirpara**, M.P. Deshpande, Growth and characterizations of tin telluride (SnTe) single crystals, *Eur. Phys. J. Plus.* 135 (2020) 1–12. <https://doi.org/10.1140/epjp/s13360-019-00022-1>.
7. A. Ghetiya, S.H. Chaki, A.J. Khimani, **A.B. Hirpara**, R.M. Kannaujiya, S. Patel, M.P. Deshpande, Growth and Characterizations of Rhenium Disulfide (ReS₂) Single Crystals, *Phys. Status Solidi Appl. Mater. Sci.* 218 (2021) 1–8. <https://doi.org/10.1002/pssa.202000687>.

8. Z.R. Parekh, S.H. Chaki, **A.B. Hirpara**, G.H. Patel, R.M. Kannaujiya, A.J. Khimani, M.P. Deshpande, CuO nanoparticles – Synthesis by wet precipitation technique and its characterization, Phys. B Condens. Matter. 610 (2021).
<https://doi.org/10.1016/j.physb.2021.412950>.
9. G.H. Patel, S.H. Chaki, R.M. Kannaujiya, Z.R. Parekh, **A.B. Hirpara**, A.J. Khimani, M.P. Deshpande, Sol-gel synthesis and thermal characterization of SnO₂ nanoparticles, Phys. B Condens. Matter. 613 (2021) 412987.
<https://doi.org/10.1016/j.physb.2021.412987>.
10. R.M. Kannaujiya, S.H. Chaki, A.J. Khimani, **A.B. Hirpara**, Z.R. Parekh, R.K. Giri, S. Patel, M.P. Deshpande, Effect of Sb doping on CVT grown SnTe single crystals electrical and thermal properties, J. Mater. Sci. Mater. Electron. 33 (2022) 20823–20836. <https://doi.org/10.1007/s10854-022-08891-6>.
11. Z.S. Kachhia, S.H. Chaki, R. Kr, Z.R. Parekh, R.M. Kannaujiya, **A.B. Hirpara**, M.P. Deshpande, J.P. Tailor, Materials Today : Proceedings Thermal decomposition study of cadmium telluride (CdTe), Mater. Today Proc. (2023).
<https://doi.org/10.1016/j.matpr.2023.02.240>
12. R.M. Kannaujiya, S.H. Chaki, A.J. Khimani, R.K. Giri, A.B. Hirpara, Y.H. Vaidya, M.P. Deshpande, Mechanistic Insights into Antibacterial and Anti-biofilm Activities against multidrug-resistant microbes of SnTe nanospheres synthesized by sonochemical method, Chem. Phys. Impact. (2023) 100219.
<https://doi.org/https://doi.org/10.1016/j.chphi.2023.100219>.
13. H.M. Mistry, M.P. Deshpande, A.B. Hirpara, N.M. Suchak, S.H. Chaki, S. Pandya, J. Valand, S. V. Bhatt, Photoresponse of surface modified graphene oxide by green synthesized silver and gold nanoparticles, Appl. Surf. Sci. 632 (2023) 157583.
<https://doi.org/10.1016/j.apsusc.2023.157583>.
14. H.M. Mistry, M.P. Deshpande, A.B. Hirpara, N.M. Suchak, S.H. Chaki, S. V. Bhatt, Bias-switchable photoconductance in surface-modified graphene oxide by green

synthesized ZnO nanoparticles, J. Alloys Compd. 969 (2023) 172328.
<https://doi.org/10.1016/j.jallcom.2023.172328>.

15. R.M. Kannaujiya, S.H. Chaki, A.J. Khimani, Z.R. Parekh, A.B. Hirpara, S.S. Soni, S.N. Bariya, M.P. Deshpande, Unveiling the optoelectronic characteristics of SnTe thin films: An extensive investigation via structural & photoresponse analysis of drop-cast deposition, Surfaces and Interfaces. 45 (2024) 103788.
<https://doi.org/10.1016/j.surfin.2023.103788>.

Poster Presentation/Conferences/Workshops/Seminars

1. Participated in the National workshop on Analytical techniques for **Material Characterization Techniques for Material Analysis (NWATMC-2017)** held at Department of Physics, Sardar Patel University, Vallabh Vidyanagar on 20th march, 2017.
2. Participated in the two days National workshop on **Advanced Characterization Techniques for Material Analysis (ACTMA-2019)** held at Department of Materials Science, Sardar Patel University, Vallabh Vidyanagar on 8th -9th march, 2019.
3. Attended **World Environment Day: Symposium and Poster Competition** organized jointly with Department of Chemistry, Sardar Patel University, K. M. Patel Institute of Physiotherapy (KMPIP), Karamsad and Voluntary Nature Conservancy (VNC).
4. Poster presented on **thermal study of direct vapour transport (DVT) grown quaternary Cu₂ZnSnS₄ single crystals** at 34th Gujarat Science Congress-2020 organized by faculty of science, Ganpat university during February 8-9,2020.
5. Paper presented on **Photocatalytic Investigation of Sonochemically Synthesized Quaternary Cu₂ZnSnS₄ Nanoparticles** in National Conference on Advanced Materials and Its Applications (NCAMA-2022) held on 5th – 6th May 2022, organized by the Department of Physics, Faculty of Arts, Science, Commerce and Management, Karpagam Academy of Higher Education, Coimbatore, Tamil Nadu, India.

6. Paper presented on **Cu₂ZnSnS₄ (CZTS) crystal-Based visible light Photodetector and its Photodetection Properties** "Innovations in Science and Technology for Industrial Applications (ICISTIA 2022)" organized by K.S. Rangasamy College of Technology, Tiruchengode, Tamil Nadu during June 1-3, 2022.
7. Paper presented on **“Biological investigation of sonochemically synthesized CZTS nanoparticles”** in the 2nd International Conference on “Sustainable Materials and Technologies for Bio and Energy Applications SMTBEA-2022”, during 13-15th, July 2022 Organized by SSN Institutions, Kalavakkam, Chennai-603110.
8. Poster presented on **Photocatalytic study of sonochemically synthesized quaternary Cu₂ZnSnS₄ nanoparticles** at National Conference on Nanomaterials (NANOPHILIA -2022) organized by Department of Science and Humanities, Providence College of Engineering during 25-08-2022 to 27-08-2022.
9. participated in the **“Basic Training Program in Nano Science and Technology** (Online mode)”, conducted by the Centre for Nano Science and Engineering, Indian Institute of Science, Bangalore, from 13 – 15 March 2023.
10. Presented a paper entitled **Photoresponse study of the dip coated Cu₂ZnSnS₄ thin film** in the 7th International Conference on Nanoscience and Nanotechnology (ICONN-2023) organized by Department of Physics and Nanotechnology, SRM IST, India during March 27- 29, 2023.

Personal Details

➤ NAME	: Anil Babubhai Hirpara
➤ FATHER’S NAME	: Babubhai Vashrambhai Hirpara
➤ MOTHER’S NAME	: Kanchanben Babubhai Hirpara
➤ DATE OF BIRTH	: 7 th Feb 1995.
➤ GENDER	: Male.
➤ NATIONALITY	: Indian.
➤ LANGUAGES KNOWN	: English, Hindi and Gujarati.
➤ HOBBIES	: Cricket, Research, Photography.

References

Prof. S. H. Chaki

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Declaration

I assure you that the above stated information is true to the best of my knowledge and I will present the documents if any required.

Yours Sincerely,

Anil B. Hirpara